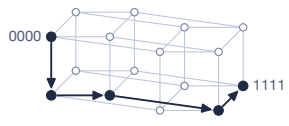
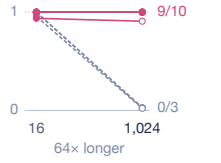
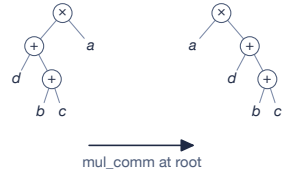
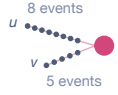
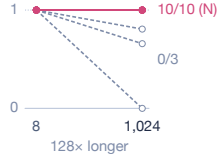
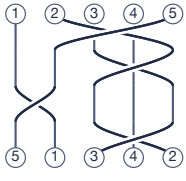
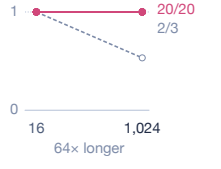
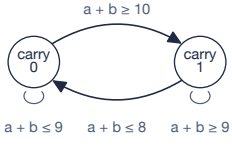
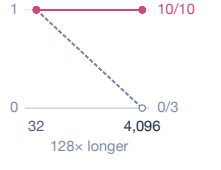
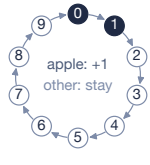
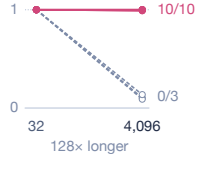
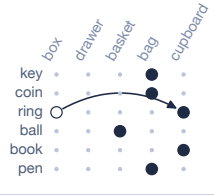
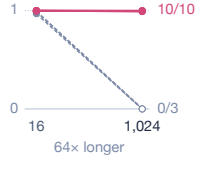
Six domains, one recipe: each is a hidden state machine read event by event, trained short and tested 64–128× longer

	domain and its state	first events, the state after each	the state machine, those events marked	a certified pair: two histories, one state	exact final state, by seed
a	program execution eight bits v0–v7, four active: 16 states; each statement walks one edge or stays	v2 = (1 - v1) % 2 ▶ 0010 v0 = (v2 + v1) % 2 ▶ 1010 v3 = (v2 + v1) % 2 ▶ 1011 v1 = 1 ▶ 1111 …		 same state: 0100 certified by: Python interpreter	
b	Lean rewrite chains the expression, one of 960 trees; each step is a lemma Lean certifies	start ▶ (d + (b + c)) × a add_comm @L ▶ ((b + c) + d) × a add_comm @L ▶ (d + (b + c)) × a mul_comm @root ▶ a × (d + (b + c)) …		 same state: a × (d + (c + b)) certified by: Lean 4 kernel	
c	S5 word problems the arrangement of 1–5, one of 120; each swap is a transposition	swap 2 5 ▶ 15342 swap 3 5 ▶ 15243 swap 1 2 ▶ 51243 swap 3 5 ▶ 51342 …		 same state: 12534 certified by: group multiplication	
d	long addition the carry: two states; the sum digit is read from a + b + carry	8 + 3 ▶ digit 1, carry 1 2 + 6 ▶ digit 9, carry 0 1 + 0 ▶ digit 1, carry 0 2 + 9 ▶ digit 1, carry 1 …		 same state: digit 9, carry 0 certified by: integer addition	
e	counting apples so far, mod 10: ten states; apples come among nine distractor words	date ▶ count 0 apple ▶ count 1 plum ▶ count 1 melon ▶ count 1 …		 same state: 7 (mod 10) certified by: counting	
f	story state where each of six objects is: 5^6 = 15,625 states	Soon after, the ring gets put in the cupboard. ▶ key: bag, coin: bag, ring: cupboard … Inside the drawer, Anna drops the coin again. ▶ key: bag, coin: drawer, ring: cupboard … Then, Anna drops the key inside the basket. ▶ key: basket, coin: drawer, ring: cupboard …		 same state: key: basket, coin: box … certified by: world simulator	

Rose: the method (readout loss, certified equivalence, centred closure, aimed contraction; (N): Lean's fixed-norm closure). Dashed grey: ordinary training. One line per seed; open end: not exact at the test length; k/n: seeds exact at the test length. n = 512 per length; addition and counting: answer exact; S5: last-event accuracy.